

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T02 · Set Notation and Venn Diagrams

Questions

Student book · questions with writing space

■ 22 pages

FAMILY MAP

Set Notation Venn Diagrams

F01	Translate and operate on written sets	7 Q 17 marks
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F02	Read or construct a two-set Venn diagram	6 Q 8 marks
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F03	Resolve a three-set Venn model	8 Q 19 marks
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Question 04

4MA1-1H Q7

MODULAR UNIT 1

5 marks

- 7 $\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$
 $A = \{\text{even numbers}\}$
 $B = \{23, 29, 31\}$
 $C = \{\text{multiples of 3}\}$

(a) List the members of the set

(i) $B \cup C$

(1)

(ii) $A' \cap C$

(1)

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes

No

☐☐

Give a reason for your answer.

(1)

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$ (c) List the members of set D

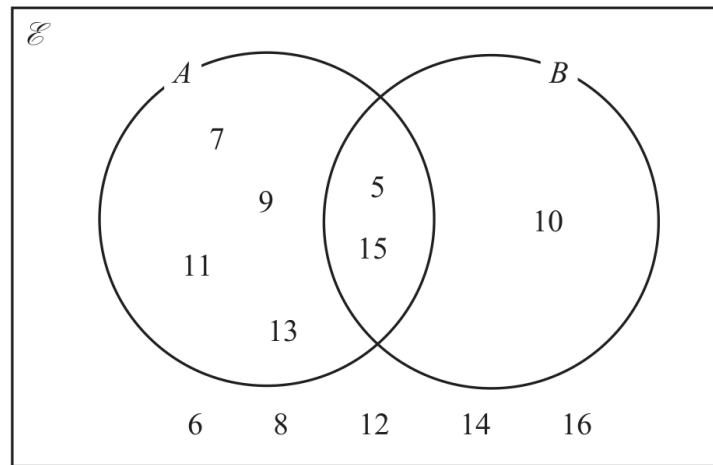
Question 01

4MA1-1H Q1 (a)

MODULAR UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(a) A

(1)

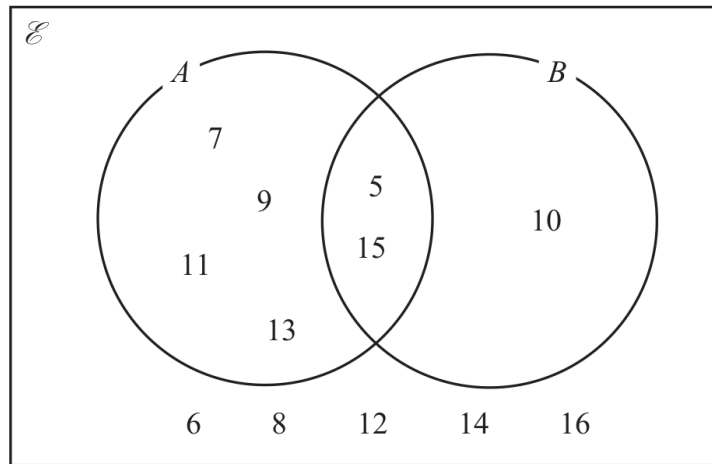
Question 02

4MA1-1H Q1 (b)

MODULAR UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(b) $A \cap B$
(1)

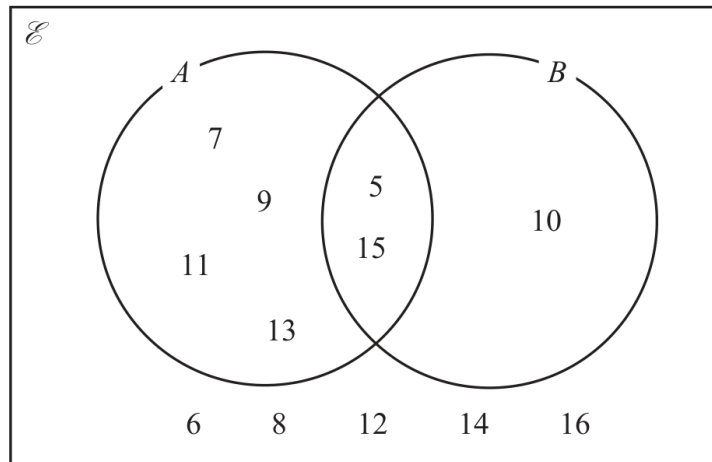
Question 03

4MA1-1H Q1 (c)

MODULAR UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(c) $(A \cup B)'$

(1)

Question 09

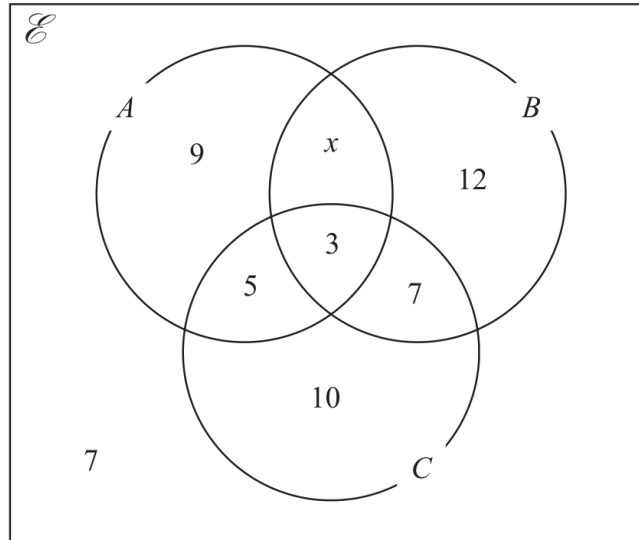
4MA1-1H Q16

MODULAR UNIT 1

3 marks

16 The Venn diagram shows a universal set $\mathcal{E}$ and sets A , B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x = \dots\dots\dots$
(1)

(b) Find $n(A')$

$\dots\dots\dots$
(1)

(c) Find $n(B' \cap C)$

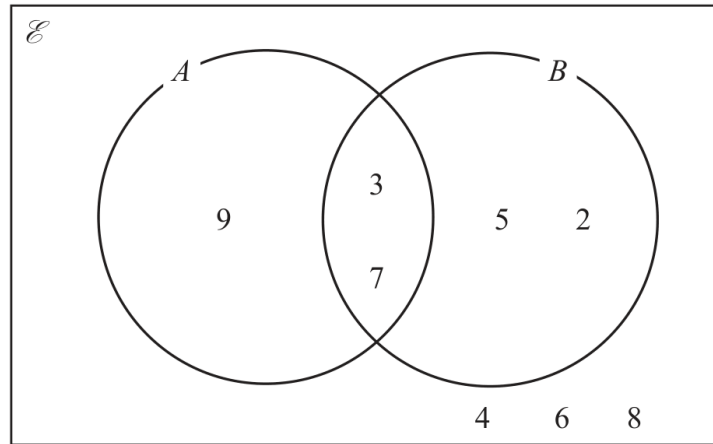
Question 04

4MA1-1H Q4 M01

MODULAR UNIT 1

1 marks

4 Here is a Venn diagram.



(a) List the members of the set B

.....

(1)

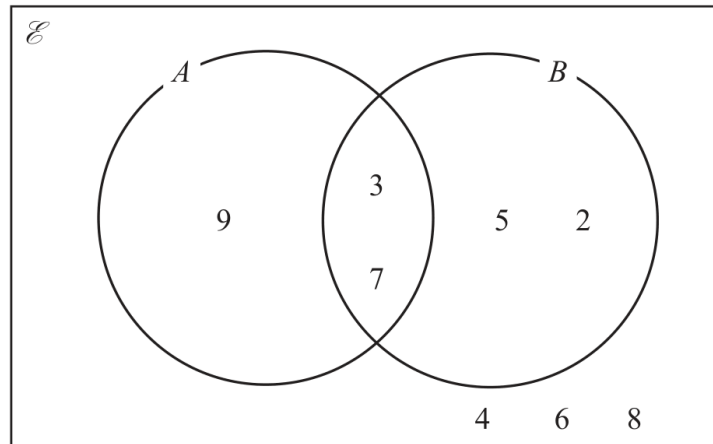
Question 05

4MA1-1H Q4 M02

MODULAR UNIT 1

1 marks

4 Here is a Venn diagram.



(b) List the members of the set $A \cap B$

.....
(1)

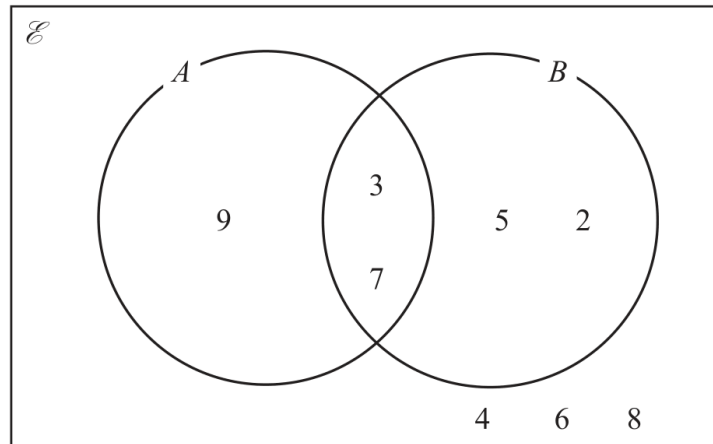
Question 06

4MA1-1H Q4 M03

MODULAR UNIT 1

1 marks

4 Here is a Venn diagram.



(c) List the members of the set A'

.....
(1)

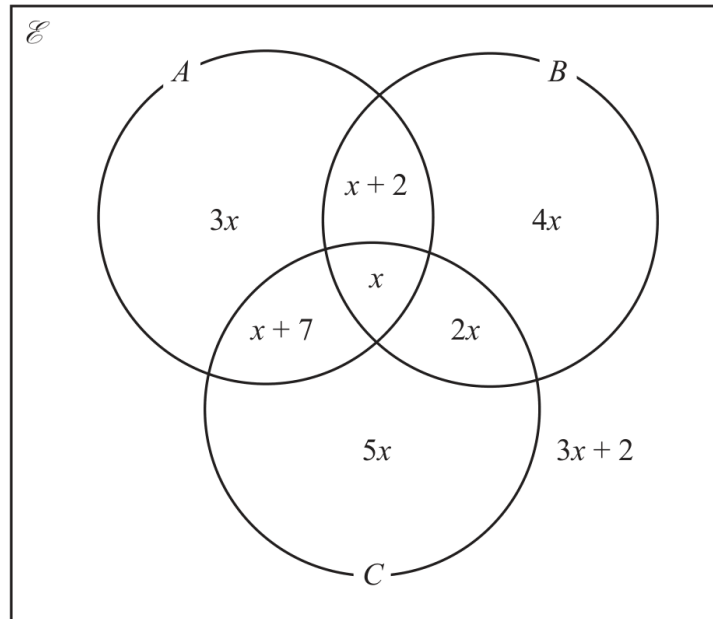
Question 26

4MA1-2H Q21 M01

MODULAR UNIT 1

4 marks

- 21 The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$n(A' \cap C) = \dots\dots\dots$

Question 06

4MA1-1H Q7

MODULAR UNIT 1

3 marks

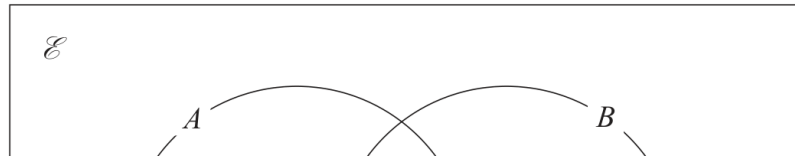
$$7 \quad \mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



$$B = \{\text{multiples of } 3\}$$

(i) $A \cap B$

January 2021 | 4MA1-1H | Question 4 | 1 marks

Chapter OL-T02: Set Notation & Venn Diagrams

(ii) $A \cup B$

16 Some students were asked the following question.

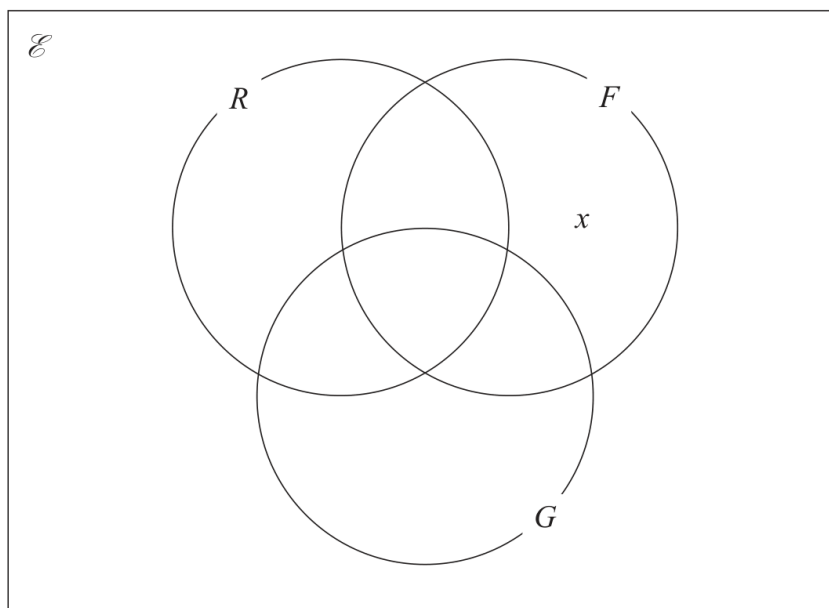
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

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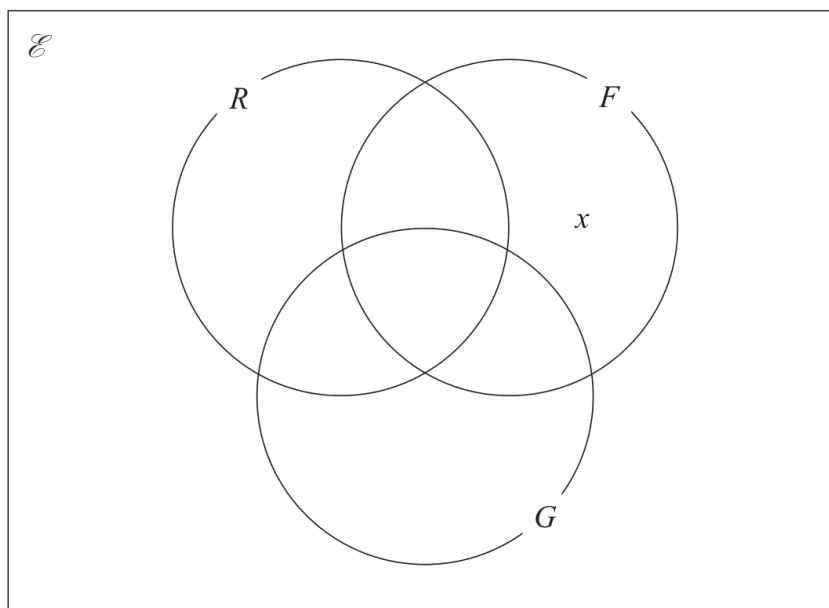
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(b) work out the number of these students that study Russian.

A Venn diagram with three overlapping circles labeled A, B, and C. The regions are numbered: 12 (A only), 9 (B only), 4 (C only), 5 (A and B), 10 (A and C), 3 (B and C), and 6 (A, B, and C). The number 8 is outside the circles.

Find

(i) $n(A \cup B)$

(ii) $n(A' \cap B')$

June 2021 | 4MA1-1H | Question 19 | 1 marks

Chapter OL-T02: Set Notation & Venn Diagrams

(iii) $n([A \cap B] \cup C)$

- 4 $\mathcal{E} = \{\text{letters of the alphabet}\}$
 $B = \{\text{b, r, a, z, i, l}\}$
 $I = \{\text{i, r, e, l, a, n, d}\}$

(a) List the members of the set

(i) $B \cup I$

(ii) $B \cap I'$

Cody writes down the statement $B \cap K = \emptyset$
 Cody's statement is wrong.

14 $\mathcal{E} = \{\text{integers } x \text{ such that } 10 \leq x \leq 25\}$

$$A = \{x : x < 18\}$$

$$B = \{x : 13 \leq x < 22\}$$

(a) Write down $n(A)$

(1)

(b) List the members of the set $(A \cup B)'$

(2)

(c) List the members of the set $A' \cap B$

(2)

$C \subset A$, $C \subset B$ and $n(C) = 5$

(d) List the members of the set C

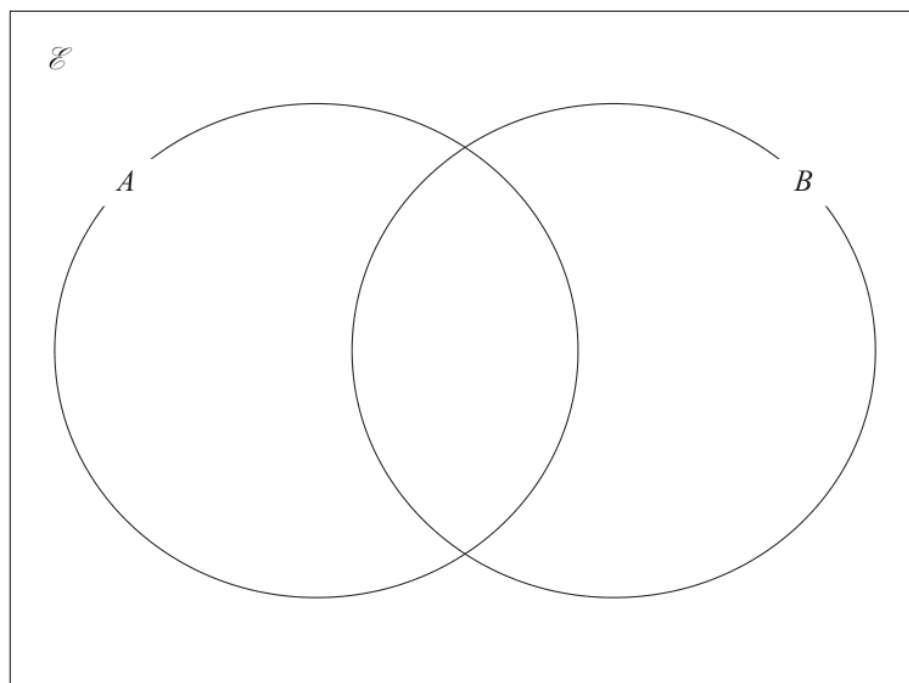
3 $\mathcal{E} = \{11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$

$A = \{\text{even numbers}\}$

$A \cap B = \{12, 16, 20\}$

$(A \cup B)' = \{17, 19\}$

Complete the Venn diagram for the sets $\mathcal{E}$, A and B



(Total for Question 3 is 3 marks)